A review of abnormal prolactin response in ME/CFS

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Abstract

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a disabling health condition, characterized by post-exertional malaise and accompanied by symptoms such as fatigue, pain, brain fog, and orthostatic intolerance. Affecting about 1 in 200 people, ME/CFS has no validated treatments and very few insights into its pathophysiology. Case-control studies of ME/CFS, or the related condition, postviral fatigue syndrome, have consistently demonstrated that patients exhibit an abnormally large increase in prolactin following administration of buspirone, a drug which acts at serotonin and dopamine receptors. Less replicated evidence of abnormal prolactin response comes from studies which tested d-fenfluramine, exercise, and insulin-induced hypoglycemia. The implications of these findings are not yet clear, though potential explanations include alterations in neurotransmitter systems or sex hormone pathways. This review discusses what is known about prolactin response in ME/CFS, and suggests future research directions.

Introduction

ME/CFS

Around 0.5-1% of the population suffers from myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) [1], a debilitating condition characterized by symptoms such as post-exertional malaise (PEM), fatigue, pain, cognitive dysfunction, and orthostatic intolerance. PEM is defined as the exacerbation or new onset of symptoms following minor physical, cognitive, emotional, or social activities, and which can persist for days or weeks, or even lead to longer term worsening. In severe cases of ME/CFS, individuals may be bedbound and require a dark and silent living environment to minimize symptoms and prevent worsening. [2].

Because ME/CFS is a heterogenous syndrome defined by symptomatology, specific definitions of ME/CFS vary, with the name "ME/CFS" itself an umbrella term that encompasses the two other most common names for this syndrome: "chronic fatigue syndrome" and "myalgic encephalomyelitis" [3]. In contrast to older ME/CFS criteria, such as those proposed by Holmes et al. [4], Sharpe et al. [5], or Fukuda et al. [6], more recent case criteria, such as the 2003 Canadian consensus criteria [7] or 2015 Institute of Medicine criteria [3] require the presence of PEM for diagnosis.

As yet, no biomarkers have been validated for ME/CFS, little is known about its pathophysiology [8], and no cure currently exists for this disorder [2]. Some of the few relatively replicated abnormalities in ME/CFS, which, if proven to be robust, may eventually provide direction for treatment development include: increased brain lactate [9–15], increased usage of the IGHV3-30 gene by B cells [16–18], and reduced natural killer cell cytotoxicity [19].

While all of the findings noted above continue to be investigated in recent years, a further ME/CFS finding, namely abnormal prolactin response to neuroendocrine challenge, was observed in numerous studies, but has not been directly tested further in over 15 years. This review aims to outline what is known about prolactin response in ME/CFS, and to suggest research directions which may provide opportunities for deeper understanding of the mechanisms underlying this condition.

Neuroendocrine challenge testing

Studies in numerous health conditions have been undertaken to investigate the functioning of the nervous system through the use of neuroendocrine challenge tests. Such tests are performed by measuring the release of pituitary hormones after administration of drugs or other interventions known to affect hormonal release. As the pituitary gland is regulated by the brain, this testing strategy can be considered a relatively non-invasive "window" into central nervous system functioning [20].

In support of the ability of these tests to allow researchers to detect altered neural functioning, researchers have demonstrated changes in prolactin response following neurotoxic exposure. For example, studies have shown changes in the prolactin response to serotonin-related drugs following serotonin neuron-damaging toxin exposure in monkeys [21] and rats [22, 23].

Prolactin

Prolactin is a protein produced within several tissues in the human body, but its main site of synthesis is the anterior pi-

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tuitary gland. This hormone is best known for its essential role in the development of mammary tissue and its ability to stimulate lactation. Additionally, prolactin exerts numerous other effects, which include the promotion of parental behavior, enhancement of immunity, modulation of lipid and glucose metabolism, and inhibition of the reproductive cycle during lactation. The most potent natural stimulant of prolactin secretion is suckling, however stress, sounds, and circadian light patterns also affect the release of this hormone [24, 25].

Unlike most hormones of the pituitary gland, prolactin is primarily regulated through inhibition instead of stimulation, whereby pituitary prolactin-secreting cells (lactotrophs) naturally secrete prolactin when hypothalamic influence is blocked. The main hormone involved in inhibiting prolactin is dopamine, which is released by neurons of the arcuate nucleus and periventricular nucleus of the hypothalamus, and which travels through the long or short portal veins to reach the lactotrophs. Additionally, several other hormones also have the ability to directly stimulate prolactin release by lactotrophs, and these include, but are not limited to, thyrotropin-releasing hormone (TRH), vasopressin, oxytocin, and vasoactive intestinal peptide (VIP) [25].

Prolactin response in ME/CFS

Buspirone

Strikingly, every single published study that has tested prolactin response to buspirone in ME/CFS has observed that patients exhibit an abnormally large increase in prolactin, which occurs about one to two hours after drug administration. The results of these studies are summarized below, as well as in Table 1.

Buspirone is marketed as a treatment for generalized anxiety disorder, though unlike classical anxiety medications such as benzodiazepines, buspirone has no activity at GABA receptors. Instead, buspirone is an agonist of 5HT1A receptors, where it exhibits strong binding affinity, and also acts as an antagonist at dopamine D2 receptors, albeit with weaker binding affinity [26].

The first study demonstrating increased prolactin response to buspirone in an ME/CFS-like condition was undertaken by Bakheit et al. in 1992. The findings from this study were reported both in Bakheit's thesis [27], as well as in a peer-reviewed paper [28]. The authors were studying postviral fatigue syndrome (PVFS), in which the criteria used for the study closely resembled CDC (1988) [4] and Oxford [5] criteria for chronic fatigue syndrome. The study criteria also required that participants' fatigue was preceded by a probable or definite viral infection, and was made worse by exercise and did not improve after bed rest. The authors analyzed males and females separately, finding, in both cases, that when compared to healthy controls or individuals with depression, those with PVFS exhibit a larger increase in plasma prolactin after administration of buspirone. Figure 1 illustrates the study results using the individual level data provided in Bakheit's thesis. Notably, and as seen in virtually all other studies of this topic, baseline prolactin was not significantly altered in the patient group.

Further evidence of increased prolactin response to buspirone was provided by John Richardson in 1995. In this study, patients who fulfilled CDC and Oxford criteria for ME/CFS were compared to family members without ME/CFS. With prolactin response defined as the ratio of prolactin levels after to before buspirone administration, the response to buspirone (50 mg, orally) in the patients was, on average, over 3 times higher than that of the controls. The study also found a correlation between prolactin response in patients and their degree of sleep disturbance [29]. A later study from Richardson and Costa again suggested very high prolactin responses to buspirone (50 mg, orally) in ME/CFS patients, however this study did not include a control group, and it is unclear if high prolactin response was a criteria for inclusion in the study [30].

In 1996, Sharpe et al. reported increased prolactin response to buspirone (0.5 mg/kg up to 45 mg, orally) in patients fulfilling Oxford criteria for ME/CFS, but who did not also exhibit symptoms of depression. Notably, plasma levels of buspirone and its metabolite, 1-(2-Pyrimidinyl)piperazine (1-PP), were not significantly different between groups after administration of buspirone, suggesting that the difference in prolactin response was not caused by differences in drug metabolism. In contrast to prolactin response, the growth hormone response to buspirone did not significantly differ between groups [31].

Also in 1996, Tahir Majeed, under the supervision of Peter O. Behan (also an author on Bakheit's 1992 PVFS paper) published a thesis testing hormonal responses to a variety of drug challenges in ME/CFS, providing further evidence of abnormal prolactin response. The study cohort included patients who fulfilled Oxford and Fukuda [6] criteria, and who had fatigue which worsened after exercise. The control group was made up of healthy, sex-matched volunteers. The ME/CFS patients exhibited a significantly larger increase in prolactin after administration of buspirone (60 mg, orally). Further findings in ME/CFS patients included normal growth hormone responses to bromocriptine and baclofen, a larger growth hormone response to pyridostigmine, smaller growth hormone responses to desipramine and dexamethasone, and a smaller adrenocorticotropic hormone (ACTH) response to ipsapirone [32].

Another 1996 paper, authored by Behan, reported increased prolactin response to buspirone (60 mg, orally) in individuals with an ME/CFS-like illness (including myalgia, fatigue worsened by exercise, and with onset after a flu-like illness) and who had previously been exposed to organophosphates [33]. Note that while this paper reports that the study included 10 male healthy controls, later court testimony from Behan in an organophosphate trial indicated that there was a mistake in the paper, and the actual control group was made up of 15 males and 15 females [34]. However, as Be-

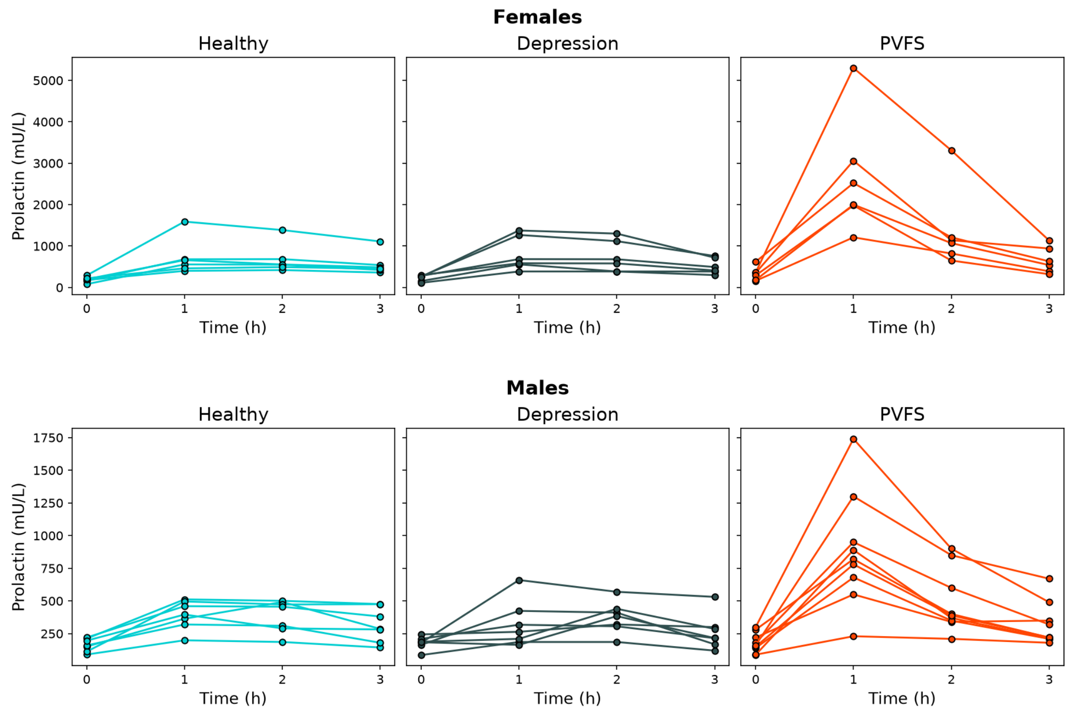

Figure 1: Prolactin response in PVFS. Plot created from data in A. M. O. Bakheit's thesis [27]. PVFS: Postviral fatigue syndrome.

han noted in court, the inclusion of females in the control group would not be expected to lead to spurious differences between groups, and would instead more likely have led to a smaller difference, as females have larger prolactin responses to buspirone than males [35].

Additionally, two studies from Racciatti et al. suggest that the authors have observed further evidence of increased prolactin response to buspirone in ME/CFS. However, one study did not include a healthy control group, and minimal detail about this finding was reported in the paper [36], while the other study was reported in an abstract which we have not been able to access [37], and with the findings briefly mentioned in a later review [38].

Finally, a 2001 case report from Sharma et al. described a female patient with ME/CFS (Oxford criteria), who exhibited an abnormally large prolactin response to buspirone (30 mg, orally). After symptom improvement, which followed a graded exercise therapy intervention, the prolactin response was measured again on two occasions, and was found to be normal both times [39].

Non-buspirone probes

Additional evidence of abnormal prolactin response was provided by studies which tested fenfluramine, exercise, and insulin-induced hypoglycemia in ME/CFS patients. Negative or conflicting results have been observed after administration of m-chlorophenylpiperazine (m-CPP) and tryptophan. Details of these studies are summarized below and in Table 2.

Several studies of prolactin response to fenfluramine in ME/CFS have been published, though with varying results. Fenfluramine is a serotonin releasing agent, and through its metabolites, agonizes dopamine and serotonin receptors [41], though the d-enantiomer formulation of fenfluramine selectively affects serotonin receptors [42]. One study tested prolactin response to 60 mg of racemic fenfluramine, with no significant difference being observed between the ME/CFS and healthy groups [43]. Three other studies have tested response to 30 mg d-fenfluramine. In two of these studies, ME/CFS patients had an abnormally increased response to d-fenfluramine [44, 45], while the third did not detect a significant difference [46].

Studies have also reported significantly *blunted* prolactin responses after neuroendocrine challenge with exercise or hypoglycemia. Bearn et al., in the same study that did not find a difference using d-fenfluramine, did detect a blunted prolactin response to insulin-induced hypoglycemia in ME/CFS patients [46]. Further, Ottenweller et al. reported a significantly smaller increase of prolactin in ME/CFS following a treadmill-based maximal exercise test [47].

Two later studies, testing other probes, have produced negative or inconsistent findings. A 2001 study did not find a significant difference between ME/CFS patients and controls in prolactin response to the serotonin agonist, m-CPP (0.25 mg/kg) [48]. Most recently, a 2010 study reported a significantly increased prolactin response to the serotonin precursor, tryptophan, in females with ME/CFS. However, this difference was not detected in females suffering from both ME/CFS and fibromyalgia, or in either of the male patient groups [49].

Table 1: Studies testing prolactin response to buspirone in ME/CFS-like conditions

Study	Participants	Probe	Prolactin response*	Notes
Bakheit, 1992 [27] (Thesis, supervised by Peter O. Behan) Bakheit et al., 1992 [28]	6F, 9M PVFS 6F, 7M depression 6F, 7M healthy	Buspirone (60 mg, orally)	PVFS: **8.2 (females)** and **5.0 (males)** Depression: 3.6 (females) and 2.1 (males) Healthy: 3.9 (females) and 2.5 (males)	Females tested during luteal phase. Only PVFS participants experienced excessive fatigue, lightheadedness, and nausea after buspirone. Minor data inconsistencies for a small number of participants, likely due to transcription errors, have been noted [40], but these are unlikely to meaningfully affect the conclusion.
Richardson, 1995 [29]	25F, 5M CFS/ME (Oxford) 30 healthy (sex ratio not reported)	Buspirone (50 mg, orally)	**CFS/ME: 4.9** Healthy: 1.8	Females tested during luteal phase. Patients experienced more nausea after buspirone compared to controls.
Sharpe et al., 1996 [31]	11M CFS (Oxford) 11M healthy	Buspirone (0.5 mg/kg up to 45 mg, orally)	**CFS: 2.9** Healthy: 2.4	Growth hormone response was not significantly different between groups. Patients had more nausea in response to buspirone but this did not correlate with prolactin response. Plasma levels of buspirone and its major metabolite, 1-PP, were not significantly different between groups.
Behan, 1996 [33]	10M CFS with previous exposure to organophosphates. 15F, 15M healthy from previous study	Buspirone (60 mg, orally)	**CFS: 4.8** Healthy: 1.9	The paper was later found to include an error: the control group was incorrectly described as 10 males. (18) Patients also had increased GH response to pyridostigmine and decreased GH response to dexamethasone.
Majeed, 1996 [32] (Thesis, supervised by Peter O. Behan)	15F, 15M CFS (Fukuda) 15F, 15M healthy	Buspirone (60 mg, orally)	**CFS: 2.8** Healthy: 2.0	Females tested during follicular phase. All participants had been medication-free for at least 3 months. No participants reported nausea after buspirone. Other studies within the thesis found: normal GH responses to bromocriptine and baclofen, larger GH response to pyridostigmine, smaller GH responses to desipramine and dexamethasone, and smaller ACTH response to ipsapirone.
Racciatti et al., 2001 [36]	3F, 2M CFS after toxic exposure 3F, 1M CFS after EBV 4F, 1M CFS with depression No healthy controls	Buspirone	"an abnormal increase of prolactine levels followed the buspirone challenge test [. . .] in all the three subgroups of patients."	Prolactin values and buspirone dosage not reported.
Sharma et al., 2001 [39]	1F CFS (Oxford)	Buspirone (30 mg, orally)	CFS: 10.9 After symptom improvement following graded exercise program: 12 months: 2.1, 18 months: 1.2	Case study.

* Prolactin response was calculated as peak average prolactin divided by average prolactin at timepoint 0. Data were approximated from plots using WebPlotDigitizer 4 when exact values were not reported.

Bold text indicates that the referenced paper reported a significant difference between the highlighted group and the healthy control group for prolactin response. Note that the definition of prolactin response for the purpose of significance testing varied by study, and was not the same as the definition used in this table.

1-PP: 1-(2-pyrimidinyl)piperazine; CFS: Chronic fatigue syndrome; EBV: Epstein-Barr Virus; F: Females; Fukuda: Fukuda et al., 1994 [6]; GH: Growth hormone; Holmes: Holmes et al., 1988 [4]; M: Males; ME: Myalgic encephalomyelitis; Oxford: Sharpe et al., 1991 [5]; PVFS: Postviral fatigue syndrome;

Table 2: Studies testing prolactin response to buspirone in ME/CFS-like conditions

Study	Participants	Probe	Prolactin response*	Notes
Cleare et al., 1995 [45]	4F, 6M CFS (Holmes and Oxford) 4F, 6M depression 8F, 12M healthy	d-fenfluramine (30 mg, orally)	**CFS: 1.5** Depression: 0.9 Healthy: 1.2	Females tested during follicular phase. Cortisol response did not differ from healthy group. When analyzing full cohort, baseline cortisol inversely correlated with prolactin response.
Bearn et al., 1995 [46]	4F, 5M CFS (Oxford and all but one fulfilled Holmes) 6F, 4M healthy	d-fenfluramine (30 mg, orally)	CFS: 1.5 Healthy: 1.7	Females tested on days 3-5 of their menstrual cycle. ACTH response was greater in patients. Cortisol response showed no group difference.
Bearn et al., 1995 [46]	4F, 5M CFS (Oxford and all but one fulfilled Holmes) 6F, 2M healthy	Insulin (0.1 or 0.15 u/kg)	**CFS: 5.1** Healthy: 6.7	Females tested on days 3-5 of their menstrual cycle. Growth hormone response trended lower in patients. Cortisol and ACTH responses were not significantly different.
Yatham et al., 1995 [43]	8F, 3M CFS (Holmes) 8F, 3M healthy	DL-fenfluramine (60 mg, orally)	CFS: 1.7 Healthy: 2.5	Cortisol response did not significantly differ between groups.
Sharpe et al., 1997 [44]	10M CFS + neurasthenia 10M healthy	d-fenfluramine (30 mg, orally)	**CFS: 1.6** Healthy: 1.1	
Ottenweller et al., 2001 [47]	17F CFS (Holmes and Fukuda) 14F healthy	Maximal exercise test on treadmill	**CFS: 1.2** Healthy: 2.2	Females tested during luteal phase. ACTH, EPI, TSH, DHPG, DOPA, and DOPAC responses were lower, GH response was higher, and NE response not significantly different.
Vassallo et al., 2001 [48]	16F, 4M neurasthenia (all but one also fulfilled Oxford and CDC criteria for CFS) 15F, 5M healthy	m-CPP (0.25 mg/kg, orally)	Average prolactin values not reported, thus ratio could not be calculated. No significant difference between groups.	Females tested during early follicular phase. Participants also underwent a placebo test which did not raise prolactin in either group.
Weaver et al., 2010 [49]	15F, 7M CFS (Fukuda) 8F, 3M CFS (Fukuda) + FM 10F, 6M healthy	L-tryptophan (120 mg/kg lean body mass)	CFS: **2.3 (females)** and 2.2 (males) CFS+FM: 2.0 (females) and 3.0 (males) Healthy: 1.9 (females) and 2.2 (males)	Females tested during late follicular phase (days 7–14).

* Prolactin response was calculated as peak average prolactin divided by average prolactin at timepoint 0. Data were approximated from plots using WebPlotDigitizer 4 when exact values were not reported.

Bold text indicates that the referenced paper reported a significant difference between the highlighted group and the healthy control group for prolactin response. Note that the definition of prolactin response for the purpose of significance testing varied by study, and was not the same as the definition used in this table.

ACTH: Adrenocorticotropic hormone; CFS: Chronic fatigue syndrome; EPI: Epinephrine; FM: Fibromyalgia; Fukuda: Fukuda et al., 1994 [6]; F: Females; Holmes: Holmes et al., 1988 [4]; GH: Growth hormone; m-CPP: m-chlorophenylpiperazine; M: Males; NE: Norepinephrine; Oxford: Sharpe et al., 1991 [5]; TSH: Thyroid-stimulating hormone;

Prolactin response in non-ME/CFS conditions

Abnormally increased prolactin response has been reported in several other conditions, such as migraine, irritable bowel syndrome (IBS), and gonadal disorders. The following section briefly outlines some examples of non-ME/CFS prolactin response research, but this should not be considered a comprehensive review of the literature.

Migraine

Abnormal prolactin response has been reported several times in migraine. Cassidy et al. have published two studies showing increased prolactin response to buspirone in migraine [50, 51]. The prolactin response to m-CPP was larger in a study that used a dosage of 0.5 mg/kg [52], but not in a study that used a smaller 0.25 mg/kg dosage of the drug [53]. A larger prolactin response was noted after administration of racemic fenfluramine [54]. Migraine patients may have an abnormally long return to baseline prolactin after administration of reserpine (seen in males) [55] or benserazide (seen in females) [56]. Studies have reported abnormally increased prolactin response in migraine patients to the D2 antagonists sulpiride [56, 57] and domperidone [57]. An abnormally large *decrease* in prolactin has been noted after administration of L-deprenyl [58]. In contrast, an abnormally *small* decrease was reported after administration of the dopamine reuptake inhibitor, nomifensine [57] and the dopamine precursor, L-DOPA [59].

Several studies have tested TRH challenges in migraine. A blunted prolactin response in migraine was seen after administration of 50 μg and 200 μg doses of TRH [60]. Another study found an increased prolactin response to TRH during migraine attacks [61]. Awaki et al. found an abnormally increased prolactin response after administration of a cocktail which included TRH, gonadotropin-releasing hormone (GnRH), and insulin [62].

Other studies of migraine did not find significant differences between groups, including after administration of domperidone [63], fenfluramine [56], lisuride [57], metaclopromide [64, 65], morphine [66], piribedil [67], and sumatriptan [68].

Gonadal disorders

Studies have also tested prolactin response in disorders related to testicular and ovarian function, with increased response being a commonly reported finding. Abnormalities in levels of hormones other than prolactin are also commonly reported in these studies.

For females with amenorrhea secondary to ovarian failure, abnormally large prolactin response to TRH has been reported both in a case report [69] and in a case-control study. In the latter study, the patients also had increased basal luteinizing hormone (LH) and follicle-stimulating hormone (FSH), decreased estrone and estradiol, and normal TSH response to TRH [70].

Similar observations of increased prolactin response have been reported in relation to testicular disorders, though findings have not been entirely consistent. In several studies from Spitz et al., males with oligo- or azoospermia were found to have larger prolactin responses to TRH [71–74] or to combined TRH and GnRH [75]. Baranowska et al. reported increased prolactin response to metoclopramide in oligo- and azoospermia [76]. Boucher et al. found increased prolactin response to combined TRH and GnRH in males with varicocele [77]. Studies have also reported blunted prolactin response to TRH in azoospermia [78], as well as normal prolactin response in oligospermia [79]. The latter study additionally did not detect an abnormal prolactin response to L-DOPA, pyridoxine, or metoclopramide [79]. A study from Winters et al. reported that early pubertal males with hypogonadotrophic hypogonadism had smaller prolactin responses to the D2 antagonist chlorpromazine [80].

Other health conditions

Studies from Dinan et al. have reported abnormally increased prolactin response to buspirone in IBS [81] and in non-ulcer dyspepsia [82–85].

In patients with major depressive disorder, the prolactin response to various probes has been observed to be blunted in some [45, 86–93], but not all studies [28, 94–101]. Additionally, one study of depression has reported *increased* prolactin response to the D2 antagonist, sulpiride [102].

Connection to sex hormones?

Females are more likely than males to be diagnosed with ME/CFS [1], which raises an intriguing possibility that the well-known influence of sex hormones on prolactin regulation, briefly discussed below, may relate to the abnormal prolactin responses observed in ME/CFS.

The clearest connection to the sex bias in ME/CFS comes from observations that females, when compared to males, tend to have larger prolactin responses to various neuroendocrine probes. Healthy females have a higher prolactin response to buspirone than males [28, 35]. Similar sex differences have been observed after administration of d-fenfluramine [103], thyrotropin-releasing hormone (TRH) [104], phenothiazine [105], and possibly chlorpromazine [106].

Evidence of sex hormone influence on prolactin regulation is additionally provided by observations that the magnitude of prolactin response to various drugs differs throughout the menstrual cycle. O'Keane et al. observed that while baseline prolactin levels in healthy females stayed relatively steady throughout the menstrual cycle, prolactin response to d-fenfluramine was highest at mid-cycle, followed by the luteal, then follicular phase, which mirrored the differing amounts of circulating estradiol measured at different phases [107]. Prolactin response to buspirone in females was found to be larger during the luteal phase than during the follicular phase or at mid-cycle [35]. Prolactin response to TRH may be somewhat larger during the follicular phase [108], though findings are inconsistent [109].

Direct evidence of sex hormone influence on prolaction secretion is provided by studies demonstrating the effect of exogenous estradiol on regulation of prolactin. TRH-induced prolactin response was seen to be substantially larger in females taking a short course of exogenous estradiol, as well as after subsequent short-term combined contraceptive use, but not in those taking long-term combined contraceptives [108]. In male rats, a prolactin response to TRH was only detectable when the rats had previously been administered estradiol [110]. Exogenous estradiol causes an increased prolactin response to ghrelin in postmenopausal women [111]. Administration of estradiol to overiectomized rats resulted in substantially larger prolactin responses to the dopamine antagonist, thioproperazine [112].

Another interesting potential connection to sex hormones, though not directly related to prolactin response, is based on Dinan et al. showing that healthy females had more symptoms of sedation in response to buspirone during the luteal phase than at other timepoints, with this also being the timepoint where prolactin response was found to be highest [35]. This may relate to observations that ME/CFS patients report more buspirone-induced side effects, such as fatigue and nausea, when compared to healthy controls [28, 29, 31].

Additionally, while increased prolactin response is a relatively rarely observed finding in other health conditions, some of the few conditions in which this has been reported multiple times also suggest connections to sex hormones. As outlined in the preceding section, migraine patients were observed in several studies to have increased prolactin response to various neuroendocrine probes [50–52, 54, 56, 57]. Like ME/CFS [1], migraine appears to also be more prevalent in females than males [113, 114]. Several studies have also reported increased prolactin response in gonadal disorders such as primary testicular failure and primary ovarian failure [70–73, 75–77]. Gonadal organs are the primary sites of sex hormone synthesis, and as such, the authors of several of the gonadal disorder studies speculate that alterations in sex hormone pathways may mediate the changes in prolactin response seen in these conditions [70, 74, 76].

An important point to keep in mind is that a hypothesis which ties estradiol to the abnormal prolactin response seen in ME/CFS would need to be explained by a mechanism which is independent of blood estradiol levels, as studies have shown that blood levels of this hormone do not appear to be significantly increased in ME/CFS [115–118]. One alternative possibility was proposed in a study of increased prolactin response to TRH in primary ovarian failure, where the authors speculated that the patients might have increased conversion of androgens to estrogens within the hypothalamus [70], based on an observation that such increased conversion was seen in castrated rabbits [119].

Future directions

The implications of abnormal prolactin response in ME/CFS are far from clear. While several of the ME/CFS prolactin response papers suggested that the findings were evidence of abnormalities in the serotonin system [28, 32, 44, 45], there is still too little direct evidence to make strong conclusions about specific pathways. Indeed, the findings of a PET study of serotonin receptors in ME/CFS suggested the opposite of what was expected based on prior prolactin response findings (the authors found down-regulation instead of upregulation of 5-HT1A receptors) [120]. Serotonin-related drugs, which were used in most of the ME/CFS studies with significant results, are unlikely to be able to directly stimulate lactotrophs [121], and may instead exert at least part of their influence on lactotrophs through altering hypothalamic dopamine release [122]. Thus, while abnormal prolactin response could be explained by changes in the receptors directly affected by the neuroendocrine probes, such as 5-HT receptors, these findings could instead be highlighting any of a number of possible downstream abnormalities, such as alterations in dopamine functioning or sensitization of lactotrophs.

A straightforward next step would be testing prolactin response to other neuroendocrine probes. For example, if dopaminergic drugs and TRH both lead to abnormal prolactin responses, this would suggest abnormally sensitized lactotrophs, as dopamine and TRH both directly affect lactotrophs [25]. On the other hand, if neither of these probes demonstrates altered responses, but responses to serotonin-related drugs are altered, this may indicate upstream abnormalities. It may be most fruitful to consider testing the probes which have already been reported to cause increased prolactin response in other health conditions. For example, in one study of migraine, the D2 receptor antagonist domperidone caused an increased prolactin response [57]. Domperidone does not appreciably cross the blood-brain barrier [123], allowing researchers to test direct lactotroph stimulation while limiting confounding due to central nervous system effects.

A finding related to levels of the prolactin regulating alpha-melanocyte-stimulating hormone (α-MSH) may also be valuable to replicate. One ME/CFS study has reported increased levels of α-MSH, albeit with substantial overlap between groups [124]. α-MSH is known to alter prolactin response, both in vitro [125, 126] and in vivo in non-human animals [127, 128]. To explore a possible connection, researchers may wish to test for a correlation between α-MSH and prolactin response in participants with ME/CFS.

One could test for a connection to sex hormones, as discussed in the previous section, by measuring the prolactin response to buspirone or another probe in ME/CFS patients and healthy controls both before and after administration of an estrogen receptor antagonist or aromatase inhibitor. If the difference between groups were found to be diminished after blocking the influence of estrogens, this would provide evidence of sex hormone involvement in the abnormal prolactin response. It may be necessary to prescribe an extended course of these drugs prior to the second prolactin challenge, as estrogens are capable of inducing long term changes in prolactin-secreting cells, for example through genomic regulation or by increasing the number of lactotrophs, as reviewed by Grattan et al. [24].

It may also be possible to explore abnormal prolactin regulation further through the use of PET scans, for example by measuring dopamine receptor D2 (DRD2) binding in the pituitary gland. If the ME/CFS abnormality does relate to sex hormones, then DRD2 levels may be altered, as is seen when estradiol is applied to anterior pituitary cells in vitro [129].

Controlling for confounders

To ensure interpretable results, and to increase the likelihood of detecting real differences, researchers should be mindful to control for relevant potential confounders when studying prolactin response in ME/CFS.

One of the most important factors to consider in studies of prolactin response is the sex of participants. As noted earlier, females demonstrate larger prolactin responses than males [28, 35, 103–106]. Thus, at minimum, groups in case-control studies should consist of the same ratio of females to males. However, it may be wise to also control for sex in an appropriate statistical model or analyze males and females separately to reduce the influence of variance associated with sex.

Additionally, as females exhibit variance in prolactin response throughout the menstrual cycle [35, 107], researchers should control for menstrual phase as well. Some of the clearest differences between ME/CFS and healthy participants occurred in studies of females tested during the luteal phase [28, 29], while a study that included females tested during the follicular phase showed a smaller group difference [32]. While one can not make strong conclusions about menstrual phase effects from such limited data, this may suggest that the optimal time for testing females would be during the luteal phase. In any case, studies of females should ensure that controls are matched for menstrual phase.

Limited evidence suggests that sedentariness may also lead to altered prolactin response. Non-endurance trained individuals, when compared to elite endurance-trained athletes, may exhibit an increased prolactin response to buspirone [130]. Cadegiani & Kater observed that non-athletes have smaller prolactin responses to insulin-induced hypoglycemia than athletes [131], which is directionally consistent with the results of an insulin study of ME/CFS [48]. Broocks et al. observed larger prolactin response to m-CPP in non-exercising subjects than in marathon runners [132], though in ME/CFS, a difference was not observed when testing a smaller dose of m-CPP [48]. In contrast, a study found no difference between endurance-trained and sedentary males in a d-fenfluramine prolactin challenge [133], and another study found that 9 weeks of exercise did not alter prolactin response to buspirone [134]. To explore the possibility that abnormal prolactin response findings in ME/CFS may be caused by deconditioning, researchers could include a control group matched for physical activity level, such as patients who are bed-bound for reasons unrelated to ME/CFS.

It may also be worthwhile to control for the effects of additional placebo-like factors on prolactin response [135]. As stress can cause changes in prolactin secretion [136, 137], this factor could potentially play a role in ME/CFS findings, for example through the stress associated with venipuncture. Although, at least one study has tested prolactin response to placebo in ME/CFS, and found that patients did not exhibit an abnormally increased prolactin response following the placebo test [48].

Further, researchers should ensure that other significant medical conditions which could influence prolactin response are not present in the study cohort. For example, liver and kidney diseases affect metabolism of buspirone [138], and this may alter prolactin response to this drug. Even if disorders of metabolism are excluded, it may still be worthwhile to test for plasma concentrations of neuroendocrine probes. Also, as some studies have observed blunted prolactin response in depression [45, 86–93], it may be beneficial to exclude participants with comorbid depression, in order to reduce the likelihood of masking an ME/CFS-associated increase in prolactin response.

Conclusion

Among the most replicated findings in ME/CFS research, abnormal prolactin response ofers the opportunity to illuminate specific dysfunctional pathways underlying this condition. However, almost all studies into this phenomenon in ME/CFS were conducted in the 1990s, with no studies directly testing prolactin response in ME/CFS in over 15 years.

Duval et al. argue that neuroendocrine challenge testing, at least in the field of psychiatry, was largely abandoned in recent years for reasons which include the difficulties of controlling for all relevant confounding influences on hormones, difficulty in recruiting drug-naive participants, and the lack of interpretability afforded by readily available but non-specific probes [139]. The literature does not appear to include evidence suggesting that prolactin-related findings in ME/CFS were false positives or caused by irrelevant confounding. One can imagine that prolactin response research in ME/CFS may have been a casualty of the pivot away from neuroendocrine research in other fields.

Considering the massive burden of ME/CFS, both in terms of prevalence as well as the substantial suffering imposed on individuals, the importance of continuing this research is clear. Through careful consideration of study design and consultation with experts in endocrinology and neurology, future research into prolactin response may uncover important insights into the pathophysiology of ME/CFS, a condition which has, so far, eluded explanation, prevention, and treatment.

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Updates

- Version 1 — August 20, 2026.
- Version 2 — August 21, 2026: Fixed citation for sentence about "increased conversion of androgens to estrogens within the hypothalamus". Minor clarifications and style improvements.
- Version 3 — August 27, 2026: Added tables, sections about confounding and neuroendocrine challenge testing, expanded conclusion, and made other minor style improvements.
- Version 4 — August 29, 2026: Added email address, improved figure, and other minor improvements.
- Version 5 — September 4, 2026: Significant update included improving structure and style, adding abstract and expanding introduction, improving figure, and adding additional relevant citations.

See all versions at https://github.com/ruvilonix/prolactin-response/.